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ENGINEERING**

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Report
ON
Autism Spectrum Disorder Prediction

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Autism Spectrum Disorder Prediction

1. Introduction

Autism Spectrum Disorder (ASD) is a neurological condition that affects communication, behavior, and social interaction. Early detection is very important for timely treatment and support. Traditional diagnosis methods are slow and require expert analysis. This project uses machine learning to predict ASD quickly and efficiently using screening data.

2. Data Description

The dataset is taken from Kaggle and contains screening data of individuals.

Features include:

- Age
- Gender
- Ethnicity
- Family history
- AQ-10 screening scores

Target Variable:

- ASD (Yes/No)

EDA Findings:

- Dataset was cleaned and preprocessed
- Categorical data encoded into numerical form
- AQ score features are strongly related to ASD
- No significant missing values

	age	gender	ethnicity	jundice	austim	contry_of_res	used_app_before	relation
0	26	f	White-European	no	no	'United States'	no	Self
1	24	m	Latino	no	yes	Brazil	no	Self
2	27	m	Latino	yes	yes	Spain	no	Parent
3	35	f	White-European	no	yes	'United States'	no	Self
4	40	f	?	no	no	Egypt	no	?
...	...	...	...	...	...	...	...	...
699	25	f	White-European	no	no	Russia	no	Self
700	34	m	Hispanic	no	no	Mexico	no	Parent
701	24	f	?	no	no	Russia	no	?
702	35	m	'South Asian'	no	no	Pakistan	no	Self
703	26	f	White-European	no	no	Cyprus	no	Self

704 rows × 8 columns

Figure 1: Class Distribution of ASD and Non-ASD Cases

Figure 1 represents the distribution of target classes in the dataset. It helps in understanding whether the dataset is balanced or imbalanced. Class distribution is important because highly imbalanced data can affect model performance and prediction accuracy. This analysis provides insight into the dataset before model training.

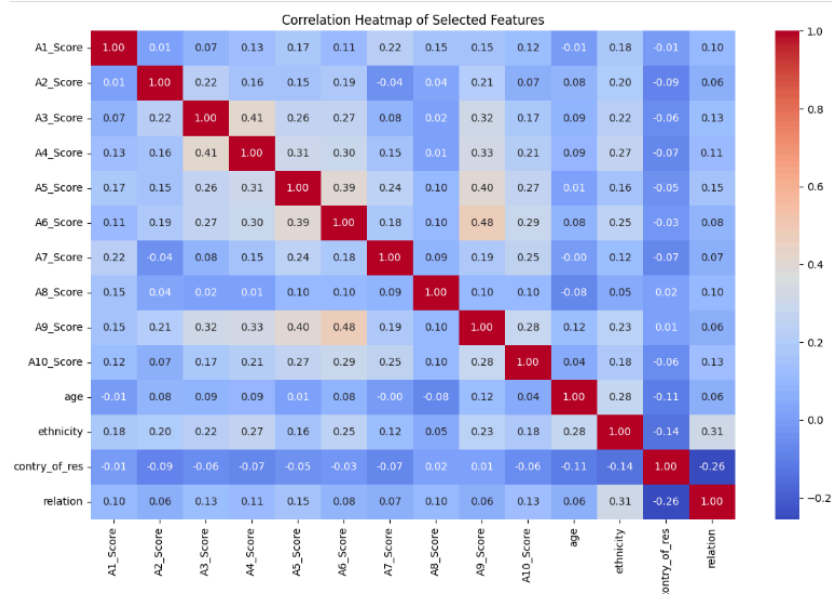


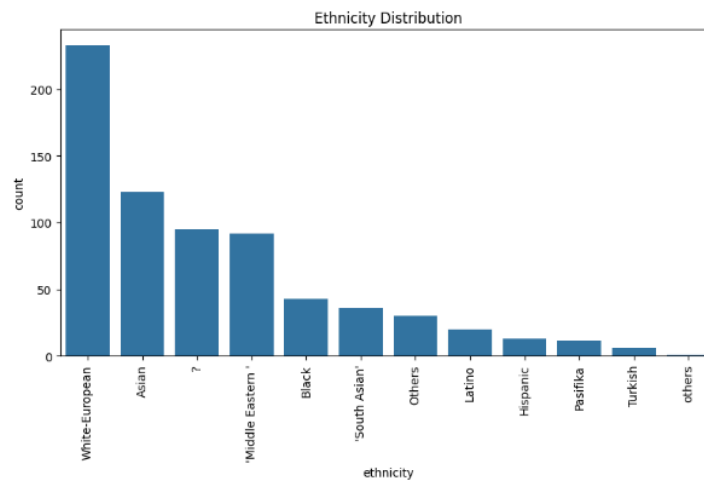
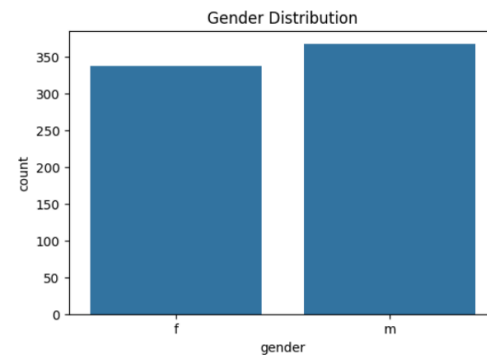
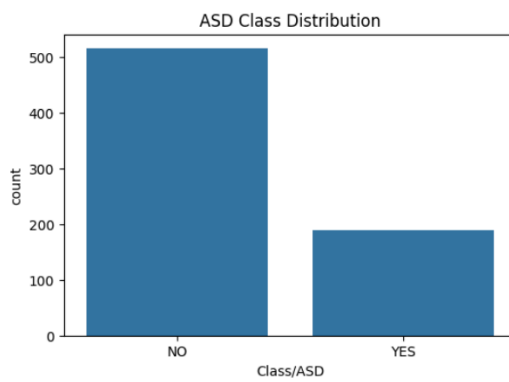
Figure 2: Correlation Heatmap of Selected Features

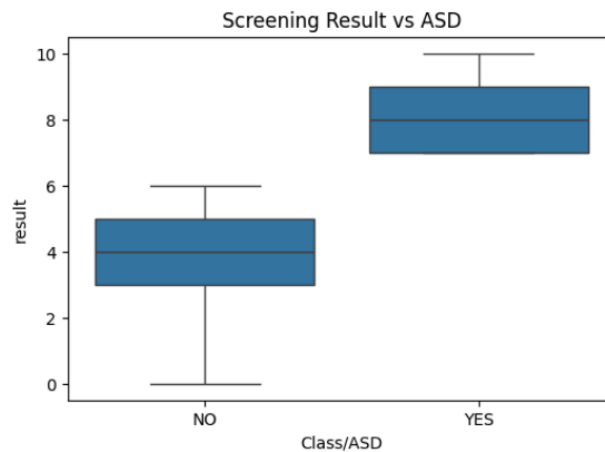
Figure 2 shows the correlation between different features in the dataset. AQ-10 screening scores show strong correlation with ASD prediction, indicating their importance. Other demographic features show weaker correlation.

3. Methodology

The process includes:

1. Data collection
2. Data preprocessing
3. Feature encoding
4. Model training
5. Model evaluation
6. Model comparison
7. Model explainability using SHAP





Flow of project:

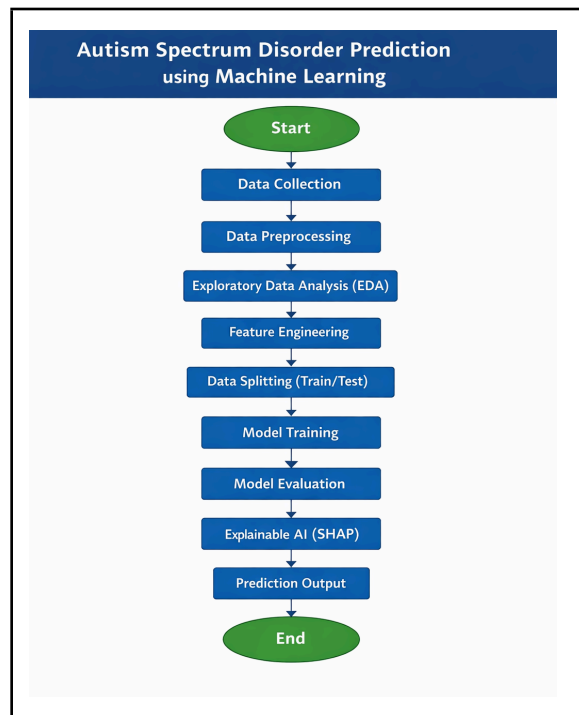


Figure 3: Workflow of Autism Spectrum Disorder Prediction System

Figure 3 shows the complete workflow followed in this project. The process starts with data collection from the Kaggle ASD dataset, followed by preprocessing, encoding, and exploratory data analysis. After feature selection, multiple machine learning models were trained and evaluated using standard performance metrics. The final step involves comparing the models to identify the best algorithm for ASD prediction.

4. Machine Learning Algorithms

- Decision Tree – Tree-based classification model
- Random Forest – Ensemble of multiple decision trees
- SVM – Finds optimal decision boundary
- KNN – Based on nearest neighbors

5. Result and Discussion

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix

ROC-AUC Curve:

ROC-AUC (Receiver Operating Characteristic – Area Under Curve) is used to evaluate the performance of classification models. It represents the trade-off between True Positive Rate and False Positive Rate.

A higher ROC-AUC value indicates better model performance. In this project, Random Forest achieved the highest ROC-AUC score of 0.99, showing excellent classification capability.

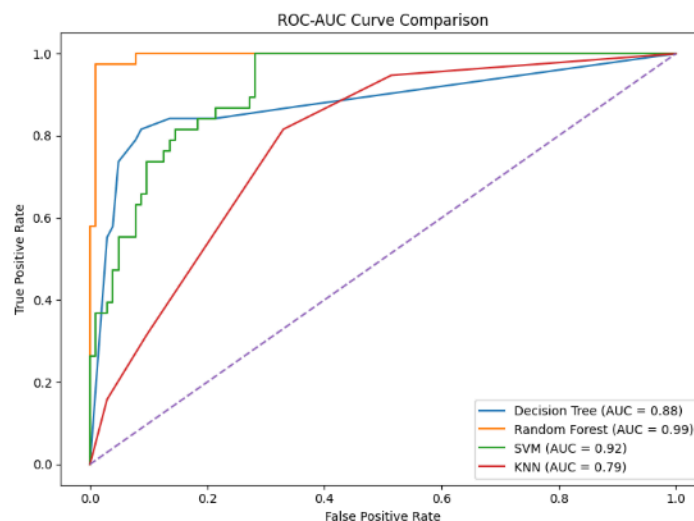


Figure 4: ROC-AUC Curve Comparison of Models

Evaluation Parameters:

Accuracy measures the overall percentage of correct predictions made by the model. Precision indicates how many cases predicted as ASD are actually ASD. Recall measures how many actual ASD cases are correctly identified by the model. F1-score is the harmonic mean of precision and recall and gives a balanced measure of model performance. In addition, the confusion matrix is used

to show true positives, true negatives, false positives, and false negatives, which helps in understanding model errors clearly.

Formulas Used:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Results :

Decision Tree – 95.74%

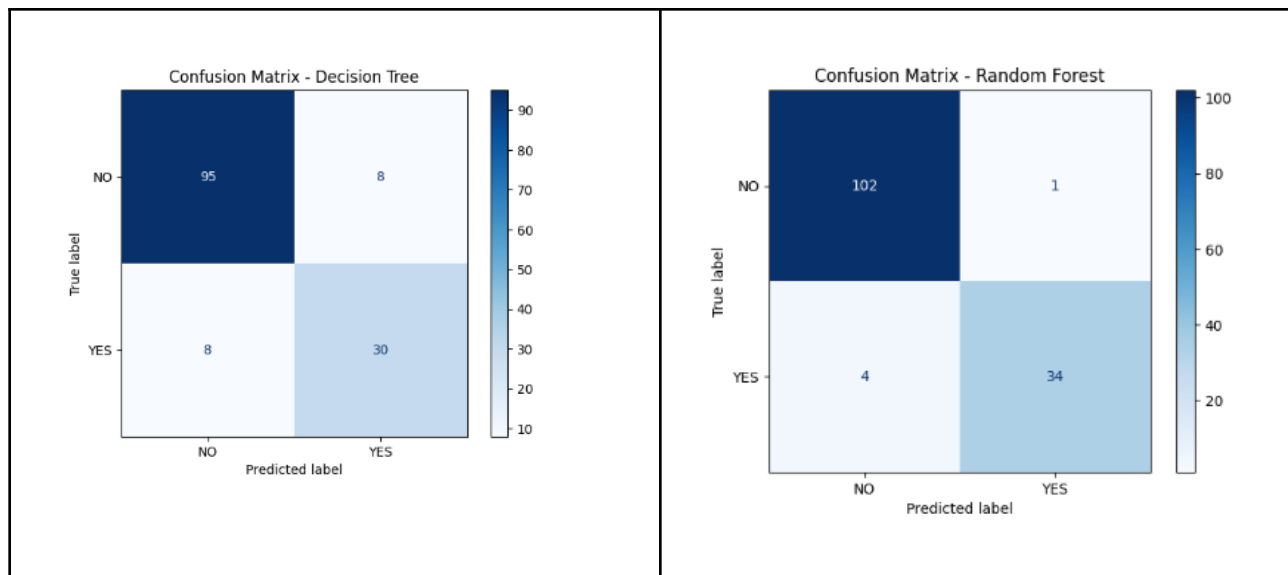
Random Forest – 96.45%

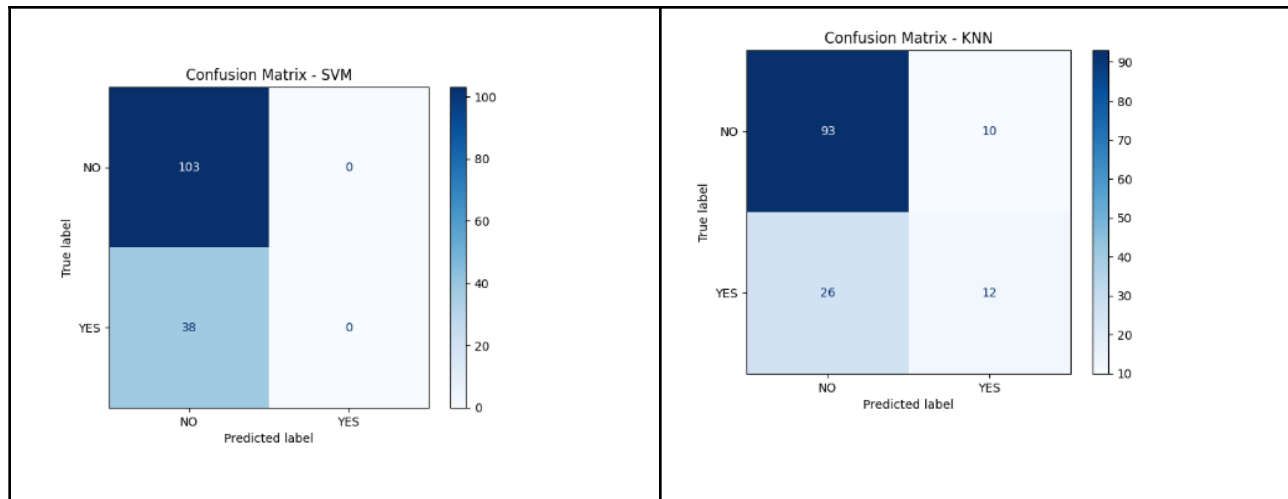
SVM – 93.0%

KNN – 85.10%

Confusion Matrix:

The confusion matrix helps in understanding the classification performance of the best model by showing the number of true positives, true negatives, false positives, and false negatives. It provides a clear view of correct and incorrect predictions and is especially useful in medical prediction tasks.





From the obtained results, it can be observed that ensemble methods performed better than basic machine learning models. Decision Tree and KNN provided moderate performance, while Random Forest achieved the best results.

Random Forest performs better because it combines multiple decision trees and reduces overfitting, resulting in higher accuracy and better generalization.

Therefore, Random Forest is the most suitable model for Autism Spectrum Disorder prediction.

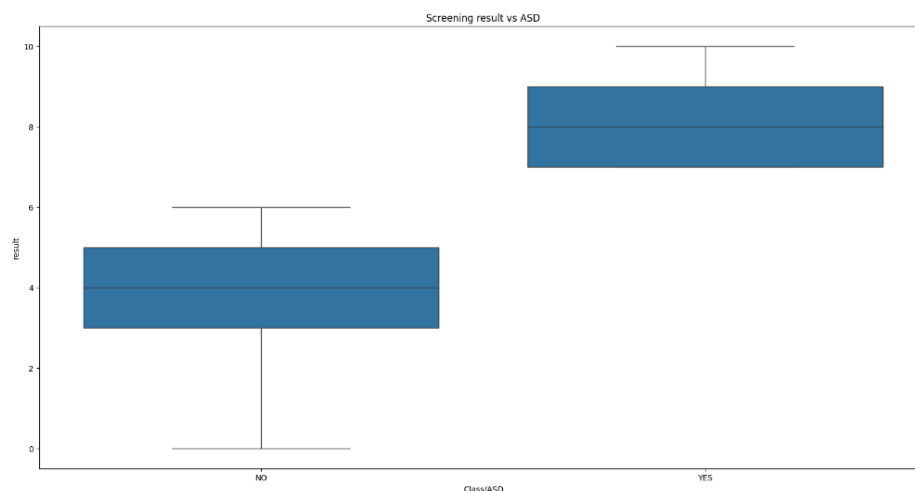


Figure 5: Accuracy Comparison of Machine Learning Models

Figure 5 compares the performance of different machine learning algorithms used in this project. Among all models, Random Forest. The graph clearly shows that ensemble methods performed

better than basic classifiers. This indicates that boosting and bagging techniques are more effective for ASD prediction.

Model Performance Comparison:

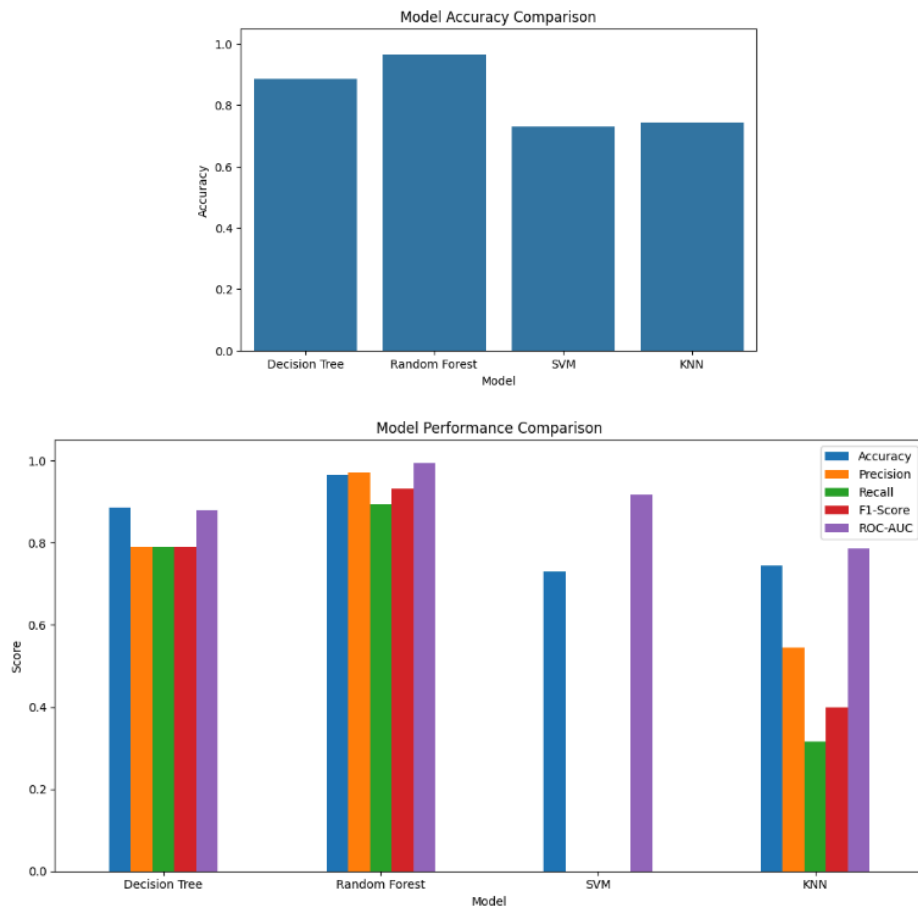


Figure 5 shows the comparison of different models based on Accuracy, Precision, Recall, and F1-score. Random Forest outperforms other models in most metrics, making it the most reliable model for ASD prediction.

Thus,

From the obtained results, it is observed that Random Forest achieved the highest accuracy of 96.45% and ROC-AUC score of 0.99.

Random Forest performs better because it is an ensemble method that combines multiple decision trees, reduces overfitting, and improves prediction performance.

Therefore, Random Forest is selected as the best model for Autism Spectrum Disorder prediction.

6. Data Source and Tools

Dataset: Kaggle ASD Dataset

<https://www.kaggle.com/code/petars1/predict-autism-spectrum-disorder/notebook>

Tools Used:

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, SHAP

AI Tool Used:

ChatGPT by OpenAI for summarization and report drafting.

7. Feature Importance

SHAP (SHapley Additive explanations) is used to interpret model predictions.

It shows how each feature contributes to the prediction.

- AQ-10 scores are the most important features
- Family history also influences prediction
- Demographic features have less impact

This improves transparency and helps understand how predictions are made.

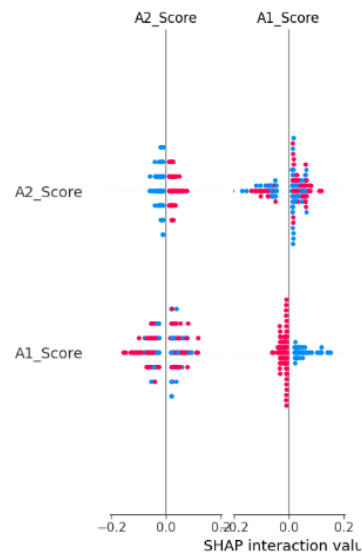
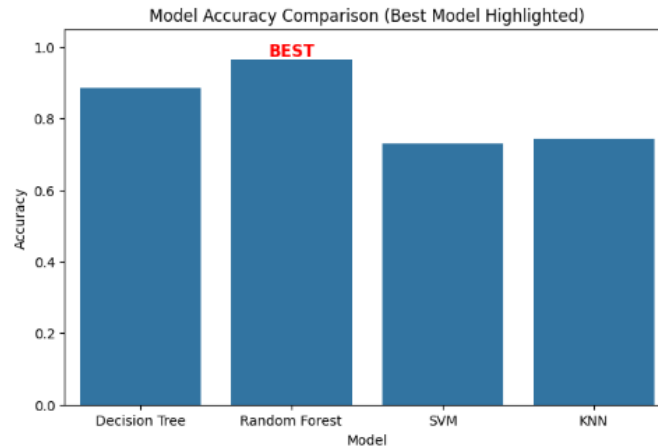


Figure 6: SHAP-Based Feature Importance for ASD Prediction

Model Accuracy Comparison:



8. Conclusion

This project demonstrates that machine learning can be effectively used for early prediction of Autism Spectrum Disorder (ASD) using screening-based data.

Different classification models such as Decision Tree, KNN, SVM, and Random Forest were applied and compared. Among all models, Random Forest achieved the highest accuracy and ROC-AUC score.

Thus, Random Forest is the most effective and reliable model for ASD prediction in this study.